CHAPTER 10

BIOMOLECULES

CHAPTER ESSENCE AT A GLANCE:
Structure and biological roles of essential carbohydrates (glucose, fructose), proteins (amino acids, peptide bonds), nucleic acids (DNA, RNA), and vitamins.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Carbohydrate Classification & Ring Structures:** Carbohydrates span monosaccharides, oligosaccharides, and polysaccharides; D-glucose exists primarily as cyclic alpha and beta pyranose hemiacetals in dynamic mutarotation equilibrium, with the open-chain form present in minute amounts.

• **Reducing Sugars & Inversion of Cane Sugar:** Reducing sugars contain free or potentially free anomeric hemiacetal groups capable of reducing Fehling's and Tollens' reagents; hydrolysis of dextrorotatory sucrose yields a levorotatory equimolar glucose-fructose mixture known as invert sugar.

• **Amino Acids & Zwitterionic Nature:** Amino acids contain both basic amino and acidic carboxyl groups, existing in neutral aqueous solution as dipolar zwitterions with characteristic isoelectric points where net electrical charge is zero.

• **Protein Architecture & Denaturation:** Polypeptides are organized into primary peptide sequences, secondary alpha-helices and beta-pleated sheets, tertiary spatial foldings, and quaternary associations; denaturation disrupts secondary and tertiary structures while leaving primary sequences intact.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Specific Rotation: } [\alpha]_D^{20} = +52.7^\circ \quad (\text{Equilibrium Mutarotation mixture of D-Glucose})$$

Mutarotation equilibrium of alpha (+112°) and beta (+19°) D-glucose anomers

$$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6\ (\text{Glucose},\ +52.5^\circ) + C_6H_{12}O_6\ (\text{Fructose},\ -92.4^\circ)$$

Inversion of cane sugar from sucrose (+66.5°) to levorotatory invert sugar (-19.9°)

$$H_3N^+\text{-CHR-COO}^- \rightleftharpoons H_2N\text{-CHR-COO}^- + H^+ \quad (\text{Zwitterion})$$

Dipolar zwitterion equilibrium of amino acid in neutral aqueous solution

$$\text{pI} = \frac{pK_{a1} + pK_{a2}}{2} \quad (\text{Isoelectric Point, Net Charge} = 0)$$

Isoelectric point equation for neutral alpha-amino acids

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** During protein denaturation, PRIMARY structure is NOT destroyed! Only secondary and tertiary hydrogen-bonded foldings are lost.

• **Alert:** [Valuation Rubric] Step 1 [Chemical Reaction or Definition]: Write chemical equation with specific reagent (e.g. Br_2 water) .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Classify carbohydrates on the basis of behavior on hydrolysis into monosaccharides, oligosaccharides, and polysaccharides. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Classification]** 1. Monosaccharides: Cannot be hydrolyzed into simpler polyhydroxy aldehydes/ketones (e.g. Glucose, Fructose, Ribose). 2. Oligosaccharides: Yield 2 to 10 monosaccharide units on hydrolysis. Disaccharides yield 2 (Sucrose, Maltose, Lactose). 3. Polysaccharides: Yield a very large number (>100) of monosaccharide units on hydrolysis (e.g. Starch, Cellulose, Glycogen). Non-sugars. [2 Marks]

• **Key Exam Result:** Monosaccharides (1 unit), Oligosaccharides (2-10), Polysaccharides (>100)

2. Write the open-chain and cyclic Haworth structures of D-glucose and explain why open-chain fails to explain certain reactions. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Open Chain Fischer]** D-Glucose is an aldohexose: $CHO-(CHOH)_4-CH_2OH$ with C-2, C-4, C-5 having -OH on right, C-3 on left. [1.5 Marks]

• **[Step 2: Haworth Cyclic Structures]** Forms a 6-membered pyranose ring by intramolecular hemiacetal formation between C-1 (-CHO) and C-5 (-OH). C-1 becomes chiral (anomeric carbon). α-D-glucopyranose: -OH at C-1 is trans to $-CH_2OH$ (drawn downwards). β-D-glucopyranose: -OH at C-1 is cis to $-CH_2OH$ (drawn upwards). They are anomers. [2.5 Marks]

• **Key Exam Result:** α and β anomers differ only in configuration at C-1 anomeric carbon

3. Write chemical equations for reaction of glucose with: (a) HI (heat), (b) Hydroxylamine, (c) Bromine water, (d) conc. HNO3, (e) Acetic anhydride. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Reactions]** (a) with HI (heat): Glucose --> n-hexane (proves all 6 carbons form an unbranched chain). (b) with NH_2OH: Glucose --> Glucose oxime (proves carbonyl >C=O group). (c) with Br_2 water: Glucose --> Gluconic acid (mild oxidation, proves carbonyl is -CHO). (d) with $(CH_3CO)_2O$: Glucose --> Glucose pentaacetate (proves five -OH groups on different carbons). (e) with conc. HNO_3: Glucose --> Saccharic acid (proves primary alcohol $-CH_2OH$ at C-6). [4 Marks]

• **Key Exam Result:** (a) n-hexane | (b) oxime | (c) gluconic acid | (d) pentaacetate | (e) saccharic acid

4. What are reducing and non-reducing sugars? Why is maltose reducing while sucrose is non-reducing? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Reducing vs Non-Reducing]** Carbohydrates that reduce Tollens' and Fehling's reagents are reducing sugars (contain free hemiacetal aldehyde/ketone group). [1.5 Marks]

• **[Step 2: Maltose vs Sucrose]** Maltose contains two α-D-glucose units joined by α(1 --> 4) glycosidic bond; C-1 of second glucose is free, making maltose reducing. In sucrose, reducing groups of both glucose (C-1) and fructose (C-2) are mutually linked via α-1,β-2 glycosidic bond; no free carbonyl exists, making sucrose non-reducing. [1.5 Marks]

• **Key Exam Result:** Maltose has free hemiacetal group; Sucrose has glycosidic linkage locking both

5. What is inversion of cane sugar? Why does the sign of optical rotation change upon hydrolysis of sucrose? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Inversion of Cane Sugar]** Sucrose is dextrorotatory (+66.5°). On hydrolysis with acid or invertase enzyme, it yields equimolar mixture of D(+)-glucose (+52.5°) and D(-)-fructose (-92.4°). [1.5 Marks]

• **[Step 2: Sign Change]** Net optical rotation of mixture = (+52.5° - 92.4°) divided by 2 = -19.95° (levorotatory). Since optical rotation inverts from dextro (+) to levo (-), hydrolysis is termed inversion and product is invert sugar. [1.5 Marks]

• **Key Exam Result:** Net rotation changes from +66.5° (dextro) to -19.9° (levo)

6. Define amino acids. Differentiate between essential and non-essential amino acids with two examples of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Amino Acids]** Compounds containing both amino ($-NH_2$) and carboxyl (-COOH) functional groups attached to α-carbon: $R-CH(NH_2)-COOH$. [1 Mark]

• **[Step 2: Essential vs Non-Essential]** Essential: Cannot be synthesized by human body and must be supplied in diet (10 amino acids, e.g. Valine, Leucine, Lysine). Non-essential: Can be synthesized by human body from other metabolites (e.g. Glycine, Alanine, Glutamic acid). [1 Mark]

• **Key Exam Result:** Essential (dietary requirement) vs Non-essential (synthesized in body)

7. What is Zwitterion structure of an amino acid? Explain its amphoteric nature and isoelectric point. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Zwitterion Structure]** In aqueous solution, dipolar ion is formed by internal proton transfer from acidic -COOH group to basic $-NH_2$ group: $H_2N-CH(R)-COOH <= => {}^{+}H_3N-CH(R)-COO^{-}$ (Zwitterion). Net electrical charge is zero. [1.5 Marks]

• **[Step 2: Amphoteric Nature]** In acidic medium (low pH), $-COO^{-}$ accepts H^{+} to become cation ${}^{+}H_3N-CH(R)-COOH$. In basic medium (high pH), ${}^{+}H_3N-$ loses H^{+} to become anion $H_2N-CH(R)-COO^{-}$. Reacts with both acids and bases. [1.5 Marks]

• **Key Exam Result:** ${}^{+}H_3N-CH(R)-COO^{-}$ is dipolar and amphoteric